

The Euler Totient (Φ) Function

A Lecture: Definitions, Formulas, Proofs, and Applications

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1 Motivation and Definition

Number theory is, in large part, the study of the multiplicative structure of the integers. One of the most natural questions we can ask about an integer n is: *how many numbers between 1 and n share no common factor with n ?* This simple counting question turns out to be extraordinarily deep, and the function that answers it — the **Euler totient function**, written $\varphi(n)$ — sits at the crossroads of elementary number theory, group theory, and modern cryptography.

Definition 1.1 (Euler's totient function). For a positive integer n , define

$$\varphi(n) = \#\{k \in \mathbb{Z} : 1 \leq k \leq n, \gcd(k, n) = 1\}.$$

That is, $\varphi(n)$ counts the integers in $\{1, 2, \dots, n\}$ that are *coprime* to n (share no prime factor with n).

In plain English: line up the numbers 1 through n . Cross out every number that shares a common factor greater than 1 with n . Whatever survives, count it — that count is $\varphi(n)$.

Example 1.2. Let $n = 12$. The numbers from 1 to 12 coprime to $12 = 2^2 \cdot 3$ are those not divisible by 2 or 3:

$$1, 5, 7, 11.$$

So $\varphi(12) = 4$.

Example 1.3. If p is prime, then *every* number from 1 to $p - 1$ is coprime to p (none of them share the factor p), so

$$\varphi(p) = p - 1.$$

This is the simplest and most important special case.

A short table of values, computed directly from the definition:

n	1	2	3	4	5	6	7	8	9	10	11	12
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4	10	4

Notice the irregularity: φ does not increase monotonically, and there is no obvious pattern from the table alone. The goal of this lecture is to tame this irregularity by deriving an exact, closed-form formula, and to show why this particular counting function is so central to algebra and cryptography.

2 Multiplicativity of φ

The single most powerful structural fact about φ is that it is *multiplicative*: it turns coprime factorizations of n into products. This is what will let us compute $\varphi(n)$ for *any* n once we know its prime factorization.

Definition 2.1. An arithmetic function $f : \mathbb{N} \rightarrow \mathbb{Z}$ (or $\mathbb{C}$) is called **multiplicative** if $f(1) = 1$ and

$$\gcd(m, n) = 1 \implies f(mn) = f(m)f(n).$$

(If the identity $f(mn) = f(m)f(n)$ holds for *all* m, n , not just coprime ones, f is called **completely multiplicative**. φ is multiplicative but *not* completely multiplicative — see Remark 2.3.)

Theorem 2.2 (Multiplicativity of φ). *If $\gcd(m, n) = 1$, then*

$$\varphi(mn) = \varphi(m)\varphi(n).$$

Proof. We use the **Chinese Remainder Theorem** (CRT). Since $\gcd(m, n) = 1$, CRT gives a bijection

$$\Psi : \mathbb{Z}/mn\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}, \quad \Psi(x \bmod mn) = (x \bmod m, x \bmod n).$$

This map is not just a bijection of sets — it is a *ring isomorphism*: it respects both addition and multiplication modulo mn . The key point for us is what happens to units (elements invertible under multiplication).

Claim. x is invertible mod mn if and only if $\Psi(x) = (x \bmod m, x \bmod n)$ has both coordinates invertible, i.e. $\gcd(x, m) = 1$ and $\gcd(x, n) = 1$.

Indeed, $\gcd(x, mn) = 1$ if and only if x shares no prime factor with m and none with n (since the prime factors of mn are exactly the union of the prime factors of m and of n , using $\gcd(m, n) = 1$), which is exactly $\gcd(x, m) = 1$ and $\gcd(x, n) = 1$.

Since Ψ is a bijection, it restricts to a bijection between the set of units mod mn and the product of the sets of units mod m and mod n :

$$(\mathbb{Z}/mn\mathbb{Z})^\times \xrightarrow{\sim} (\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times.$$

Taking cardinalities of both sides (a bijection preserves size, and the size of a product set is the product of sizes):

$$\varphi(mn) = |(\mathbb{Z}/mn\mathbb{Z})^\times| = |(\mathbb{Z}/m\mathbb{Z})^\times| \cdot |(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(m)\varphi(n). \quad \square$$

Remark 2.3. φ is *not* completely multiplicative: for example $\varphi(2 \cdot 2) = \varphi(4) = 2$, but $\varphi(2)\varphi(2) = 1 \cdot 1 = 1 \neq 2$. The hypothesis $\gcd(m, n) = 1$ in Theorem 2.2 is essential — it is exactly what makes the CRT bijection work.

Remark 2.4 (Why CRT is the right tool — intuition). The whole argument is really this one idea: knowing a residue mod mn is *the same information* as knowing a residue mod m together with a residue mod n , when m, n are coprime. And being “coprime to mn ” is a purely local condition — you just need to avoid the prime factors of m and separately avoid the prime factors of n . Splitting the problem along coprime lines lets you count independently and multiply.

3 The Value of φ at a Prime Power

Multiplicativity reduces the computation of $\varphi(n)$ to computing $\varphi(p^a)$ for a prime power. This is easy to do directly.

Proposition 3.1. *For a prime p and integer $a \geq 1$,*

$$\varphi(p^a) = p^a - p^{a-1} = p^{a-1}(p - 1) = p^a \left(1 - \frac{1}{p}\right).$$

Proof. An integer k with $1 \leq k \leq p^a$ fails to be coprime to p^a exactly when $p \mid k$ (since the only prime factor of p^a is p). The multiples of p in the range $[1, p^a]$ are

$$p, 2p, 3p, \dots, p^{a-1} \cdot p,$$

and there are exactly p^{a-1} of them. Hence the number of integers in $[1, p^a]$ *coprime* to p^a is the total count minus the multiples of p :

$$\varphi(p^a) = p^a - p^{a-1}. \quad \square$$

In plain English: out of every p consecutive integers, exactly one is divisible by p and the rest are not. So the “surviving fraction” is $1 - \frac{1}{p}$, and this proposition simply applies that fraction to the block $[1, p^a]$.

Example 3.2. $\varphi(9) = \varphi(3^2) = 3^2 - 3^1 = 9 - 3 = 6$. Check directly: coprime numbers in $\{1, \dots, 9\}$ to 9 are 1, 2, 4, 5, 7, 8 — six of them. Correct.

$$\varphi(32) = \varphi(2^5) = 2^5 - 2^4 = 32 - 16 = 16.$$

4 The Product Formula

Combining Theorem 2.2 (multiplicativity) with Proposition 3.1 (prime powers) gives the master formula.

Theorem 4.1 (Euler's product formula). *Let $n \geq 1$ have prime factorization*

$$n = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}.$$

Then

$$\varphi(n) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right) = \prod_{i=1}^r p_i^{a_i-1} (p_i - 1).$$

Proof. Since the prime powers $p_1^{a_1}, \dots, p_r^{a_r}$ are pairwise coprime, repeated application of Theorem 2.2 gives

$$\varphi(n) = \varphi(p_1^{a_1}) \varphi(p_2^{a_2}) \cdots \varphi(p_r^{a_r}).$$

By Proposition 3.1, $\varphi(p_i^{a_i}) = p_i^{a_i} \left(1 - \frac{1}{p_i}\right)$. Substituting,

$$\varphi(n) = \prod_{i=1}^r p_i^{a_i} \left(1 - \frac{1}{p_i}\right) = \left(\prod_{i=1}^r p_i^{a_i}\right) \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right) = n \prod_{i=1}^r \left(1 - \frac{1}{p_i}\right). \quad \square$$

In plain English: start with all n numbers. For each distinct prime p_i dividing n , throw away the fraction $1/p_i$ of numbers divisible by p_i . Because divisibility by different primes behaves “independently” in this counting sense (this independence is exactly what multiplicativity encodes), you can just multiply the surviving fractions together.

Example 4.2. Compute $\varphi(360)$. First factor: $360 = 2^3 \cdot 3^2 \cdot 5$.

$$\varphi(360) = 360 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) = 360 \cdot \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{4}{5}.$$

Computing: $360 \cdot \frac{1}{2} = 180$; $180 \cdot \frac{2}{3} = 120$; $120 \cdot \frac{4}{5} = 96$. So $\varphi(360) = 96$.

Example 4.3. Compute $\varphi(1000)$. Factor: $1000 = 2^3 \cdot 5^3$.

$$\varphi(1000) = 2^2(2-1) \cdot 5^2(5-1) = 4 \cdot 1 \cdot 25 \cdot 4 = 400.$$

5 Gauss's Divisor-Sum Identity

The next classical fact reveals φ as the building block of the identity function under divisor sums.

Theorem 5.1 (Gauss). *For every positive integer n ,*

$$\sum_{d|n} \varphi(d) = n,$$

where the sum runs over all positive divisors d of n .

Proof. Partition the set $\{1, 2, \dots, n\}$ according to the value of $\gcd(k, n)$. For each divisor d of n , let

$$S_d = \{k : 1 \leq k \leq n, \gcd(k, n) = d\}.$$

Every $k \in \{1, \dots, n\}$ has $\gcd(k, n)$ equal to *some* divisor of n , and this value is unique, so the sets S_d (as d ranges over divisors of n) partition $\{1, \dots, n\}$:

$$n = \sum_{d|n} |S_d|.$$

Now we compute $|S_d|$. If $\gcd(k, n) = d$, write $k = d \cdot m$. Then $\gcd(k, n) = d$ becomes $\gcd(dm, n) = d$, i.e. (dividing through by d , using $d \mid n$) $\gcd(m, n/d) = 1$. Also $1 \leq k \leq n$ becomes $1 \leq m \leq n/d$. So

$$S_d \longleftrightarrow \{m : 1 \leq m \leq n/d, \gcd(m, n/d) = 1\}$$

is a bijection ($k \mapsto k/d$), and the right-hand set has size exactly $\varphi(n/d)$ by definition. Hence $|S_d| = \varphi(n/d)$, and

$$n = \sum_{d|n} \varphi(n/d) = \sum_{d|n} \varphi(d),$$

where the last equality holds because as d ranges over all divisors of n , so does n/d (they are paired up, just relabeled). $\square$

In plain English: sort the numbers 1 through n into bins by “how much of n they share” (their $\gcd$ with n). Each bin, after rescaling, is itself a totient-counting problem for a smaller number. Adding up all the bins recovers everything, i.e. all of n .

Example 5.2. Let $n = 12$, whose divisors are 1, 2, 3, 4, 6, 12.

$$\varphi(1) + \varphi(2) + \varphi(3) + \varphi(4) + \varphi(6) + \varphi(12) = 1 + 1 + 2 + 2 + 2 + 4 = 12. \checkmark$$

Remark 5.3. In the language of Dirichlet convolution, Theorem 5.1 says $\varphi * \mathbf{1} = \text{Id}$, where $\mathbf{1}(n) = 1$ for all n and $\text{Id}(n) = n$. By Möbius inversion, this can be inverted to recover

$$\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d} = n \sum_{d|n} \frac{\mu(d)}{d},$$

where μ is the Möbius function. Expanding this sum over the prime factors of n reproduces exactly the product formula of Theorem 4.1 — a second, independent derivation of the same result.

6 Euler's Theorem and Fermat's Little Theorem

This is the theorem that gives φ its name recognition outside pure number theory — it is the engine behind modular exponentiation and RSA cryptography.

Theorem 6.1 (Euler's Theorem). *If $\gcd(a, n) = 1$, then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Proof. Let $U = \{r_1, r_2, \dots, r_{\varphi(n)}\}$ be the set of residues modulo n that are coprime to n (a *reduced residue system*); by definition $|U| = \varphi(n)$.

Consider multiplying every element of U by a modulo n :

$$ar_1, ar_2, \dots, ar_{\varphi(n)} \pmod{n}.$$

Step 1: each $ar_i \pmod{n}$ still lies in U . Since $\gcd(a, n) = 1$ and $\gcd(r_i, n) = 1$, and the gcd of a product with n depends only on prime factors, $\gcd(ar_i, n) = 1$. So $ar_i \pmod{n}$ is again coprime to n , i.e. it is (the residue of) some element of U .

Step 2: the map $r_i \mapsto ar_i \pmod{n}$ is injective on U . Suppose $ar_i \equiv ar_j \pmod{n}$. Since $\gcd(a, n) = 1$, a has a multiplicative inverse $a^{-1} \pmod{n}$; multiplying both sides by a^{-1} gives $r_i \equiv r_j \pmod{n}$, so $r_i = r_j$ (as elements of the residue system).

An injective map from a finite set U to itself is a bijection. Therefore multiplying by a merely *permutes* the elements of U :

$$\{ar_1 \pmod{n}, \dots, ar_{\varphi(n)} \pmod{n}\} = \{r_1, \dots, r_{\varphi(n)}\} \quad \text{as sets.}$$

Step 3: multiply everything together. Since both sides are the same set, their products $\pmod{n}$ agree:

$$\prod_{i=1}^{\varphi(n)} (ar_i) \equiv \prod_{i=1}^{\varphi(n)} r_i \pmod{n}.$$

The left side is $a^{\varphi(n)} \prod_i r_i$. So

$$a^{\varphi(n)} \prod_{i=1}^{\varphi(n)} r_i \equiv \prod_{i=1}^{\varphi(n)} r_i \pmod{n}.$$

Since each r_i is coprime to n , so is their product $R := \prod r_i$, hence R is invertible mod n . Multiplying both sides by R^{-1} :

$$a^{\varphi(n)} \equiv 1 \pmod{n}. \quad \square$$

In plain English: the numbers coprime to n form a closed little universe under multiplication mod n — multiplying all of them by a fixed a (itself coprime to n) just shuffles them around. Since shuffling doesn't change the product of the whole set, the extra factor of $a^{\varphi(n)}$ that appeared must equal 1.

Corollary 6.2 (Fermat's Little Theorem). *If p is prime and $p \nmid a$, then*

$$a^{p-1} \equiv 1 \pmod{p}.$$

Equivalently, for every integer a , $a^p \equiv a \pmod{p}$.

Proof. Take $n = p$ in Theorem 6.1. Since p is prime, $\varphi(p) = p - 1$ (as $\gcd(a, p) = 1 \iff p \nmid a$), giving $a^{p-1} \equiv 1 \pmod{p}$. Multiplying both sides by a gives $a^p \equiv a \pmod{p}$, and this last form also trivially holds when $p \mid a$ (both sides are 0), so it holds for all a . $\square$

Example 6.3. Compute $7^{222} \pmod{15}$. Since $\gcd(7, 15) = 1$ and $\varphi(15) = \varphi(3)\varphi(5) = 2 \cdot 4 = 8$, Euler's theorem gives $7^8 \equiv 1 \pmod{15}$. Write $222 = 8 \cdot 27 + 6$, so

$$7^{222} = (7^8)^{27} \cdot 7^6 \equiv 1^{27} \cdot 7^6 = 7^6 \pmod{15}.$$

Now $7^2 = 49 \equiv 4$, $7^4 \equiv 4^2 = 16 \equiv 1$, so $7^6 = 7^4 \cdot 7^2 \equiv 1 \cdot 4 = 4 \pmod{15}$. Hence $7^{222} \equiv 4 \pmod{15}$. This is exactly the kind of computation that makes modular exponentiation with huge exponents tractable.

7 The Group-Theoretic Viewpoint: $(\mathbb{Z}/n\mathbb{Z})^\times$

Everything above can — and should — be repackaged in the language of groups, because that is where φ truly lives.

Definition 7.1. The set of residues modulo n that are coprime to n , together with multiplication mod n , forms a group:

$$(\mathbb{Z}/n\mathbb{Z})^\times = \{[k] \in \mathbb{Z}/n\mathbb{Z} : \gcd(k, n) = 1\}.$$

This is called the **multiplicative group of units** modulo n .

Proposition 7.2. $(\mathbb{Z}/n\mathbb{Z})^\times$ is indeed a group under multiplication mod n , and

$$|(\mathbb{Z}/n\mathbb{Z})^\times| = \varphi(n).$$

Sketch. Closure: if $\gcd(a, n) = \gcd(b, n) = 1$ then $\gcd(ab, n) = 1$ (no common prime factors are introduced), so $(\mathbb{Z}/n\mathbb{Z})^\times$ is closed under multiplication. *Identity:* $[1]$ is coprime to n and acts as identity. *Associativity:* inherited from $\mathbb{Z}$. *Inverses:* if $\gcd(a, n) = 1$, Bézout's identity gives integers x, y with $ax + ny = 1$, so $ax \equiv 1 \pmod{n}$; thus $[x]$ is a multiplicative inverse of $[a]$, and $\gcd(x, n) = 1$ too (else $[x]$ couldn't be invertible), so $[x] \in (\mathbb{Z}/n\mathbb{Z})^\times$. The cardinality equals $\varphi(n)$ by the very definition of φ . $\square$

Theorem 7.3 (Euler's Theorem, restated via Lagrange). *Theorem 6.1 is exactly the statement that in the finite group $(\mathbb{Z}/n\mathbb{Z})^\times$ of order $\varphi(n)$, every element raised to the power of the group order equals the identity.*

Proof. This is an instance of **Lagrange's theorem**: if G is a finite group and $g \in G$, the order of the cyclic subgroup $\langle g \rangle$ divides $|G|$, say $|G| = k \cdot \text{ord}(g)$. Then $g^{|G|} = (g^{\text{ord}(g)})^k = e^k = e$. Applying this with $G = (\mathbb{Z}/n\mathbb{Z})^\times$ (order $\varphi(n)$) and $g = [a]$ recovers $a^{\varphi(n)} \equiv 1 \pmod{n}$ instantly, without repeating the permutation argument of Section 6 by hand — that argument *is*, in essence, a proof of Lagrange's theorem specialized to this group. $\square$

This reframing is the deep reason $\varphi(n)$ shows up everywhere: it is not merely a counting function, it is *the order of a specific, ubiquitous finite group*. Any structural fact about finite groups (Lagrange's theorem, structure theorems for abelian groups, existence of elements of certain orders) becomes a fact about $(\mathbb{Z}/n\mathbb{Z})^\times$, and hence a fact about modular arithmetic.

Theorem 7.4 (Structure of $(\mathbb{Z}/n\mathbb{Z})^\times$). *The group $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic if and only if*

$$n \in \{1, 2, 4, p^k, 2p^k\}$$

*for an odd prime p and $k \geq 1$. When $(\mathbb{Z}/n\mathbb{Z})^\times$ is cyclic, a generator is called a **primitive root** modulo n .*

Remarks on proof. The full proof (due essentially to Gauss) proceeds by: (i) proving $(\mathbb{Z}/p^k\mathbb{Z})^\times$ is cyclic of order $\varphi(p^k)$ for odd primes p by lifting a primitive root mod p via Hensel-type lifting; (ii) showing $(\mathbb{Z}/2\mathbb{Z})^\times$ and $(\mathbb{Z}/4\mathbb{Z})^\times$ are cyclic but $(\mathbb{Z}/2^k\mathbb{Z})^\times$ for $k \geq 3$ is *not* cyclic — it is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{k-2}\mathbb{Z}$; (iii) using the CRT isomorphism of Theorem 2.2 to show a product of two or more cyclic groups of even order cannot be cyclic, which rules out all other composite forms of n . We omit the full details here as they constitute a lecture of their own, but the shape of the argument is a beautiful illustration of how φ 's multiplicativity (Theorem 2.2) controls the *group structure*, not just the group's size. $\square$

Example 7.5. $(\mathbb{Z}/5\mathbb{Z})^\times = \{1, 2, 3, 4\}$ has order $\varphi(5) = 4$ and is cyclic, generated by 2: $2^1 = 2$, $2^2 = 4$, $2^3 = 3$, $2^4 = 1$. So 2 is a primitive root mod 5.

By contrast, $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ has order $\varphi(8) = 4$ but is *not* cyclic: every element satisfies $x^2 \equiv 1 \pmod{8}$ (check: $3^2 = 9 \equiv 1$, $5^2 = 25 \equiv 1$, $7^2 = 49 \equiv 1$), so it is isomorphic to the Klein four-group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, not $\mathbb{Z}/4\mathbb{Z}$. This matches the theorem: $8 = 2^3$ is not in $\{1, 2, 4, p^k, 2p^k\}$.

8 Application: RSA Cryptography

The single most famous real-world use of φ is in the **RSA public-key cryptosystem**.

- **Key generation.** Choose two large primes p, q and set $n = pq$. Compute $\varphi(n) = (p-1)(q-1)$ using Theorem 4.1. Choose a public exponent e with $\gcd(e, \varphi(n)) = 1$, and compute the private exponent d as the inverse of e modulo $\varphi(n)$: $ed \equiv 1 \pmod{\varphi(n)}$.
- **Encryption.** A message m (with $\gcd(m, n) = 1$, generically true) is encrypted as $c \equiv m^e \pmod{n}$.
- **Decryption.** The recipient computes $c^d \pmod{n}$. Correctness follows directly from Euler's theorem:

$$c^d \equiv (m^e)^d = m^{ed} = m^{1+k\varphi(n)} = m \cdot (m^{\varphi(n)})^k \equiv m \cdot 1^k = m \pmod{n},$$

using $ed = 1 + k\varphi(n)$ for some integer k (from $ed \equiv 1 \pmod{\varphi(n)}$), and Theorem 6.1 to collapse $m^{\varphi(n)} \equiv 1$.

- **Security.** An attacker who only knows n (not p, q) cannot easily compute $\varphi(n)$, because doing so is computationally equivalent to factoring n — and factoring large numbers is believed to be computationally infeasible. This is precisely why $\varphi(n)$ must be kept secret even though n itself is public.

In plain English: $\varphi(n)$ is the “clock size” for exponents in modular arithmetic mod n — exponents can be reduced mod $\varphi(n)$ without changing the result (Euler's theorem), exactly the way angles can be reduced mod 360. RSA hides this clock size as the actual secret that protects the message.

9 Further Properties, Bounds, and Asymptotics

Proposition 9.1 (Parity and simple bounds). *For $n > 2$, $\varphi(n)$ is even. Moreover $\varphi(n) \geq \sqrt{n}$ for all $n \geq 1$ except $n = 2$, and $\varphi(n) \leq n - 1$ with equality iff n is prime.*

Sketch. Evenness: for $n > 2$, the map $k \mapsto n - k$ pairs up elements of $\{1, \dots, n - 1\}$ coprime to n into disjoint pairs $\{k, n - k\}$ (they are distinct since n is not $2k$ when... more carefully: $\gcd(k, n) = 1 \iff \gcd(n - k, n) = 1$, and $k = n - k$ would force $n = 2k$, i.e. $2 \mid n$ and $k = n/2$, but then $\gcd(k, n) = k > 1$ for $n > 2$, a contradiction), so the coprime residues split into pairs, giving an even count. *Upper bound:* immediate from the definition, since n itself is never coprime to n (for $n > 1$), and equality $\varphi(n) = n - 1$ forces *every* smaller positive integer to be coprime to n , which happens iff n is prime. $\square$

Theorem 9.2 (Average order of φ).

$$\sum_{n \leq x} \varphi(n) = \frac{3}{\pi^2} x^2 + O(x \log x).$$

Equivalently, the average value of $\varphi(n)/n$ over $n \leq x$ tends to $3/\pi^2 = 6/\pi^2 \approx 0.6079$ as $x \rightarrow \infty$.

Idea of proof. Using Möbius inversion, $\varphi(n) = \sum_{d|n} \mu(d) \frac{n}{d}$. Summing over $n \leq x$ and swapping the order of summation:

$$\sum_{n \leq x} \varphi(n) = \sum_{d \leq x} \mu(d) \sum_{\substack{n \leq x \\ d|n}} \frac{n}{d} = \sum_{d \leq x} \mu(d) \sum_{m \leq x/d} m = \sum_{d \leq x} \mu(d) \left(\frac{1}{2} \left(\frac{x}{d} \right)^2 + O\left(\frac{x}{d} \right) \right).$$

The main term becomes $\frac{x^2}{2} \sum_{d \leq x} \frac{\mu(d)}{d^2} \approx \frac{x^2}{2} \cdot \frac{1}{\zeta(2)} = \frac{x^2}{2} \cdot \frac{6}{\pi^2} = \frac{3x^2}{\pi^2}$, using $\sum_{d=1}^{\infty} \mu(d)/d^2 = 1/\zeta(2) = 6/\pi^2$. The error terms are controlled to give the stated $O(x \log x)$ bound. $\square$

In plain English: although $\varphi(n)$ fluctuates wildly from one n to the next (it is large when n is prime, comparatively small when n has many small prime factors), *on average* a randomly chosen number below x has about 60.8% of the integers below it coprime to it. The constant $6/\pi^2$ is the same constant that gives the probability that two random integers are coprime — not a coincidence, since both facts trace back to the density of squarefree numbers and the Euler product for $\zeta(2)$.

Proposition 9.3 (Euler product / Dirichlet series). *For $\Re(s) > 2$,*

$$\sum_{n=1}^{\infty} \frac{\varphi(n)}{n^s} = \frac{\zeta(s-1)}{\zeta(s)}.$$

Sketch. Because φ is multiplicative, its Dirichlet series factors as an Euler product over primes:

$$\sum_{n=1}^{\infty} \frac{\varphi(n)}{n^s} = \prod_p \left(1 + \frac{\varphi(p)}{p^s} + \frac{\varphi(p^2)}{p^{2s}} + \dots \right) = \prod_p \left(1 + \frac{p-1}{p^s} + \frac{p(p-1)}{p^{2s}} + \dots \right).$$

Summing the geometric-type series inside each factor (using $\varphi(p^k) = p^{k-1}(p-1)$) simplifies each Euler factor to $\frac{1-p^{-s}}{1-p^{1-s}}$, and taking the product over all primes gives $\zeta(s-1)/\zeta(s)$, since $\zeta(s) = \prod_p (1-p^{-s})^{-1}$. $\square$

10 Summary of Key Formulas

Statement	Formula
Definition	$\varphi(n) = \#\{1 \leq k \leq n : \gcd(k, n) = 1\}$
Prime value	$\varphi(p) = p - 1$
Prime power	$\varphi(p^a) = p^{a-1}(p - 1) = p^a(1 - 1/p)$
Multiplicativity	$\varphi(mn) = \varphi(m)\varphi(n)$ if $\gcd(m, n) = 1$
Product formula	$\varphi(n) = n \prod_{p n} \left(1 - \frac{1}{p}\right)$
Gauss's identity	$\sum_{d n} \varphi(d) = n$
Möbius form	$\varphi(n) = n \sum_{d n} \frac{\mu(d)}{d}$
Euler's theorem	$a^{\varphi(n)} \equiv 1 \pmod{n}$, if $\gcd(a, n) = 1$
Fermat's little theorem	$a^{p-1} \equiv 1 \pmod{p}$, if $p \nmid a$ (special case $n = p$)
Group order	$\varphi(n) = (\mathbb{Z}/n\mathbb{Z})^\times $
Average order	$\sum_{n \leq x} \varphi(n) \sim \frac{3}{\pi^2} x^2$
Dirichlet series	$\sum_{n \geq 1} \frac{\varphi(n)}{n^s} = \frac{\zeta(s-1)}{\zeta(s)}$

11 Exercises

- (1) Compute $\varphi(2025)$, $\varphi(1001)$, and $\varphi(2^{10})$.
- (2) Prove that $\varphi(n)$ is a power of 2 if and only if n is of the form $2^a p_1 p_2 \cdots p_k$ where the p_i are distinct Fermat primes. (This fact is exactly the condition, due to Gauss, for a regular n -gon to be constructible with straightedge and compass.)
- (3) Show directly (without Theorem 5.1) that $\sum_{d|n} \varphi(d) = n$ for $n = p^a$, by summing the formula of Proposition 3.1 over $d = p^0, p^1, \dots, p^a$ and checking it telescopes to p^a .
- (4) Use Euler's theorem to compute $3^{1000} \bmod 14$.
- (5) Prove that if $\gcd(a, n) = 1$, the order of a in $(\mathbb{Z}/n\mathbb{Z})^\times$ (the least $k > 0$ with $a^k \equiv 1 \pmod{n}$) always divides $\varphi(n)$. (Hint: Lagrange's theorem, or adapt the division-algorithm argument used to prove Lagrange's theorem for cyclic subgroups.)
- (6) Find all n with $\varphi(n) = 12$.